

PAULA MAGDY

Software Engineer

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PROFILE

Full-Stack Software Engineer with production experience across TypeScript (Next.js/NestJS), PHP (Laravel), and Python (Django/Flask) on AWS, Azure, and Huawei Cloud. AWS Certified Solutions Architect contributing across data modeling, API design, CI/CD, monitoring, and security, with a track record of measurable gains in API performance and deployment speed.

PROFESSIONAL EXPERIENCE

Challenge Group | Full-Stack Software Engineer

Jun 2025 - Present

Cairo, Egypt • AWS/Huawei Cloud partner delivering production platforms for government and enterprise clients across Egypt, KSA and the UAE.

Contributed to multiple production platforms as part of cross-functional Agile delivery teams:

- Developed domain-driven NestJS modules and Laravel APIs over PostgreSQL/Prisma and MySQL for government and enterprise platforms serving 5,000+ concurrent users, and cut API latency by ~35% under production load through targeted indexing and query tuning.
- Moved deployments to a container-first CI/CD pipeline (Docker → ECR → ECS via CodeBuild/CodePipeline), cutting deployment time from ~20 minutes to under 5.
- Built the request-and-bid marketplace flow with escalation handling and an in-app wallet on Paymob, letting users pay and withdraw funds without leaving the platform.
- Delivered real-time and messaging features, including WebSocket chat (Laravel Reverb), voice/SMS via Twilio, FCM push notifications, and S3-backed media storage, giving users live chat, calls, and alerts in Arabic and English.
- Hardened API layers across multiple products with Sanctum/JWT authentication, RBAC, attribute-based routing, and self-validating DTOs, rejecting unauthorized and malformed requests before they reached business logic.
- Provisioned and maintained AWS infrastructure (VPC, EC2, RDS, S3, CloudFront, IAM, Secrets Manager) for 5+ production environments, and independently designed and ran the Azure API Management layer.
- Instrumented services with CloudWatch monitoring and structured logging, and maintained automated test coverage (Pest, PHPUnit, Jest, Vitest) that runs in CI to catch regressions before release.
- Built Next.js/React interfaces for onboarding and results-entry flows, role-based admin panels (Filament/Livewire), and live-updating leaderboards on a shared, responsive component set reused across products.
- Independently designed, built, and shipped a serverless Amazon Lex chatbot on AWS Lambda, integrated with Twilio, automating customer-support conversations and cutting the manual triage load on the support team.

Stack: NestJS, Laravel, Next.js, React, TypeScript, PHP, Python, PostgreSQL, Prisma, MySQL, Redis, Filament (Livewire), Sanctum, Reverb (WebSockets), Twilio, Paymob, Firebase Cloud Messaging (FCM), Docker, AWS (ECS, ECR, Lambda, Lex, RDS, S3, CloudFront, CloudWatch, Secrets Manager), Microsoft Azure, Azure API Management, CI/CD, Pest, PHPUnit.

Information Technology Institute (ITI) | Full-Stack Engineer

Oct 2024 - Jun 2025

Cairo, Egypt • Government-sponsored intensive software-engineering program (~1,200 hours)

- Built full-stack applications in Node.js/Express and Python (Django, Flask) with React frontends, secure REST APIs, structured authentication flows, and role-based access control.
- Modeled both relational (SQL) and NoSQL databases, applying normalization and indexing to keep queries efficient as datasets grew.
- Containerized services with Docker and deployed to AWS, applying system-design fundamentals and working in Agile sprints from design through delivery.

Freelance | Front-End Developer

Oct 2023 - Oct 2024

Cairo, Egypt • Front-end work for multiple clients, mainly e-commerce websites, from full storefront builds to feature and maintenance tasks.

- Built and shipped production-ready React storefronts for several clients, each a responsive, cross-device UI wired to REST APIs for the product catalog and cart.
- Set up reusable component libraries with centralized state via React's Context API, removing prop-drilling and keeping catalog and cart state consistent as product ranges and traffic grew.
- Optimized rendering and asset loading on new builds and existing client sites, cutting page-load times and keeping large product listings smooth as catalogs scaled.

SKILLS

Languages: TypeScript, JavaScript, Python, PHP, SQL, C++

Frontend: React.js, Next.js, Tailwind CSS, Bootstrap, CSS, HTML, Livewire, Filament, Blade

Backend: Node.js, NestJS, Express.js, Laravel, Django, Flask, REST APIs, WebSockets, Reverb, Serverless, JWT, RBAC

Databases & ORMs: PostgreSQL, MySQL, MongoDB, Redis, Prisma ORM, Eloquent ORM, Query Optimization

Cloud & DevOps: AWS, Huawei Cloud, Microsoft Azure, Docker, Kubernetes, CI/CD

Testing & Practices: Vitest, Jest, PHPUnit, Pest, React Testing Library, Playwright, Domain-Driven Design, Agile/Scrum

CERTIFICATIONS

Huawei HCCDP – Solution Architectures

Jun 2026

Professional-tier certification in end-to-end solution design on Huawei Cloud, focused on high availability, security, and cost efficiency across the platform's compute (ECS), storage (OBS), networking (VPC/ELB), and managed-database services.

AWS Certified Solutions Architect – Associate (SAA-C03)

Jun 2026

Validates the design of fault-tolerant, scalable, and secure AWS architectures across VPC networking, EC2/Lambda compute, S3 and RDS/DynamoDB data layers, IAM, and multi-AZ high availability and disaster recovery.

Huawei HCCDA – Tech Essentials

May 2026

Associate-level foundation in Huawei Cloud covering core compute, storage, and networking services, application deployment, and the platform's developer toolchain.

AWS Certified Cloud Practitioner

Apr 2026

Foundational AWS certification spanning cloud concepts and economics, the shared-responsibility security model, pricing and billing, and the platform's core service categories.

EDUCATION

B.Sc., Computer Science and Information Systems

2018 – 2022

October 6 University, Giza, Egypt

- Relevant coursework: Algorithms, Data Structures, Cloud Computing, Database Systems, Software Engineering.
- Graduation project (grade A): Learning Management System (Node.js, Express, MongoDB).